

lab10

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Background

The main repository of high-resolution structural data on biomolecules is called **Protein Data Bank** (PDB).

PDB Statistics

What is in the PDB in terms of molecular determination

```
pdb <- read.csv("Data Export Summary.csv")
head(pdb)
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	217,375
2	1	0	14,283
3	0	0	16,277
4	3	1	5,004
5	0	0	244
6	0	4	22

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

This print out above is `pdb$X.ray` is “character” not “numeric”. Two functions can help here are `sub()` and `as.numeric()`:

```
sum(as.numeric(sub(",", "", pdb$X.ray)))
```

```
[1] 203794
```

Or we can make a function:

```
rm.comma <- function(x) {
  sum(as.numeric(sub(",", "", x)))
}
```

```
n.xray <- rm.comma(pdb$X.ray)
n.em <- rm.comma(pdb$EM)
n.tot <- rm.comma(pdb$Total)

n.xray/n.tot*100
```

```
[1] 80.48577
```

```
n.em/n.tot*100
```

```
[1] 13.38046
```

We could also use a different import function for this CSV.

```
library(readr)
pdb <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
-- Column specification -----
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(pdb)
```

```
# A tibble: 6 x 9
  `Molecular Type`    `X-ray`    EM    NMR Integrative `Multiple methods` Neutron
  <chr>             <dbl> <dbl> <dbl>      <dbl>          <dbl>    <dbl>
1 Protein (only)    180758 23111 12813      348            229      84
2 Protein/Oligosacch~ 10488 3741  34      8             11       1
3 Protein/NA         9205 6751  287     26             8        0
4 Nucleic acid (only) 3154  250 1578     3            15       3
5 Other              178   27   35     4             0        0
6 Oligosaccharide (o~  11    0    6     0             1        0
# i 2 more variables: Other <dbl>, Total <dbl>
```

```
n.xray <- sum(pdb$`X-ray`)
n.em <- sum(pdb$EM)
n.tot <- sum(pdb$Total)
```

```
n.xray / n.tot * 100
```

```
[1] 80.48577
```

```
n.em / n.tot * 100
```

```
[1] 13.38046
```

Q2: What proportion of structures in the PDB are protein?

```
pdb$Total[1]
```

```
[1] 217375
```

The total number of protein sequences in Uniprot is 202,556,314

```
217375/202556314*100
```

```
[1] 0.1073158
```

Keypoint: WE have a very small structural coverage from known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography).

```
protein_rows <- grepl("Protein", pdb$`Molecular Type`)  
  
n.protein <- sum(pdb$Total[protein_rows])  
n.total <- sum(pdb$Total)  
  
n.protein / n.total * 100
```

```
[1] 97.91868
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

4998

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Water molecules appear as one atom because X-ray crystallography usually only detects the oxygen atom clearly. The two hydrogen atoms are too small/low in electron density, so they are usually not shown in the PDB model.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

The critical conserved water molecule is usually water 301. In the structure, its residue number is 301, often labeled HOH 301 or WAT 301.

Visualizing PDB data with Mol-star

Main stand alone web version with all feature is at <https://molstar.org/viewer/>.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

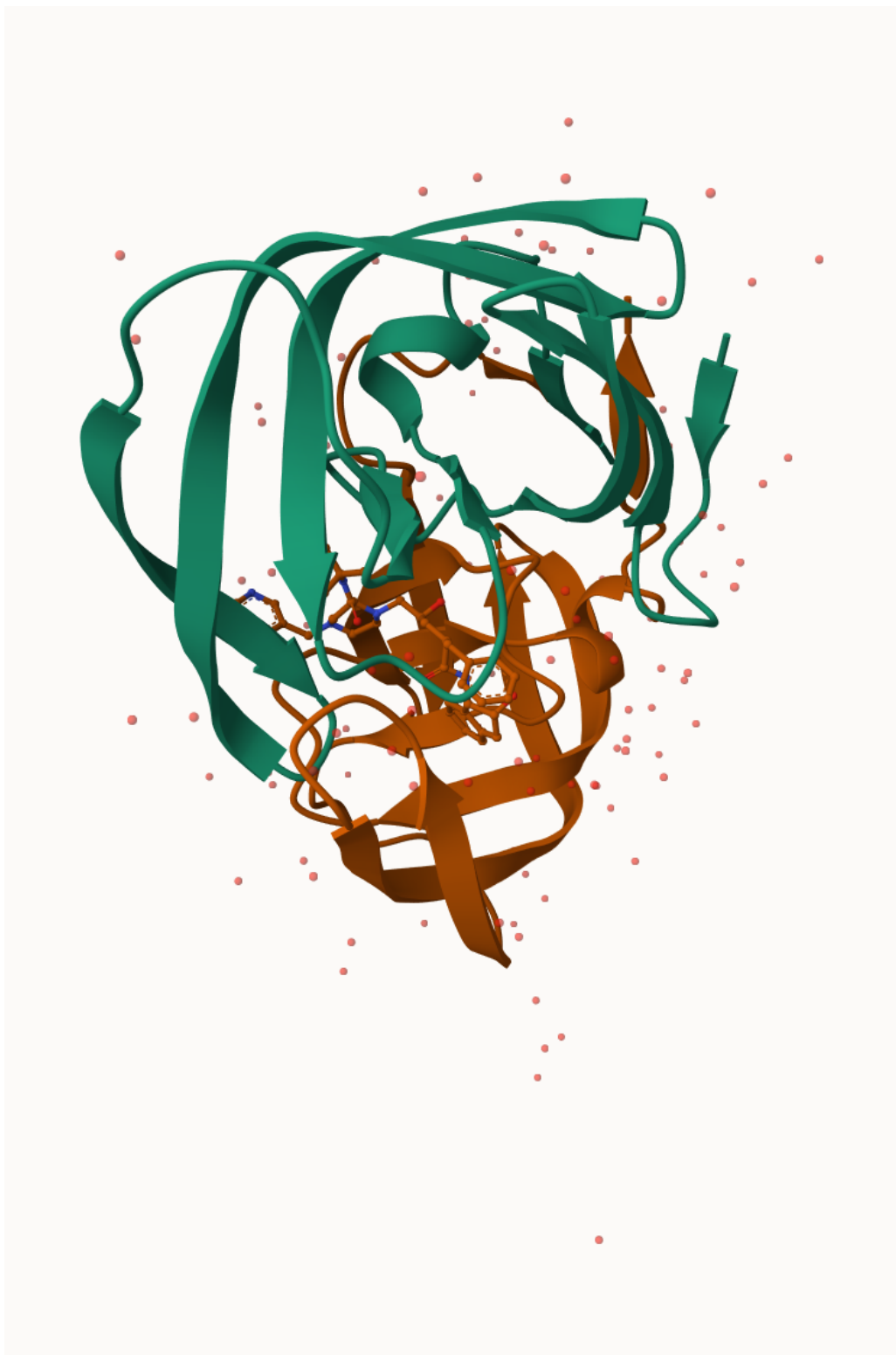


Figure 1: The HIV-protease enzyme is a homodimer with two changes

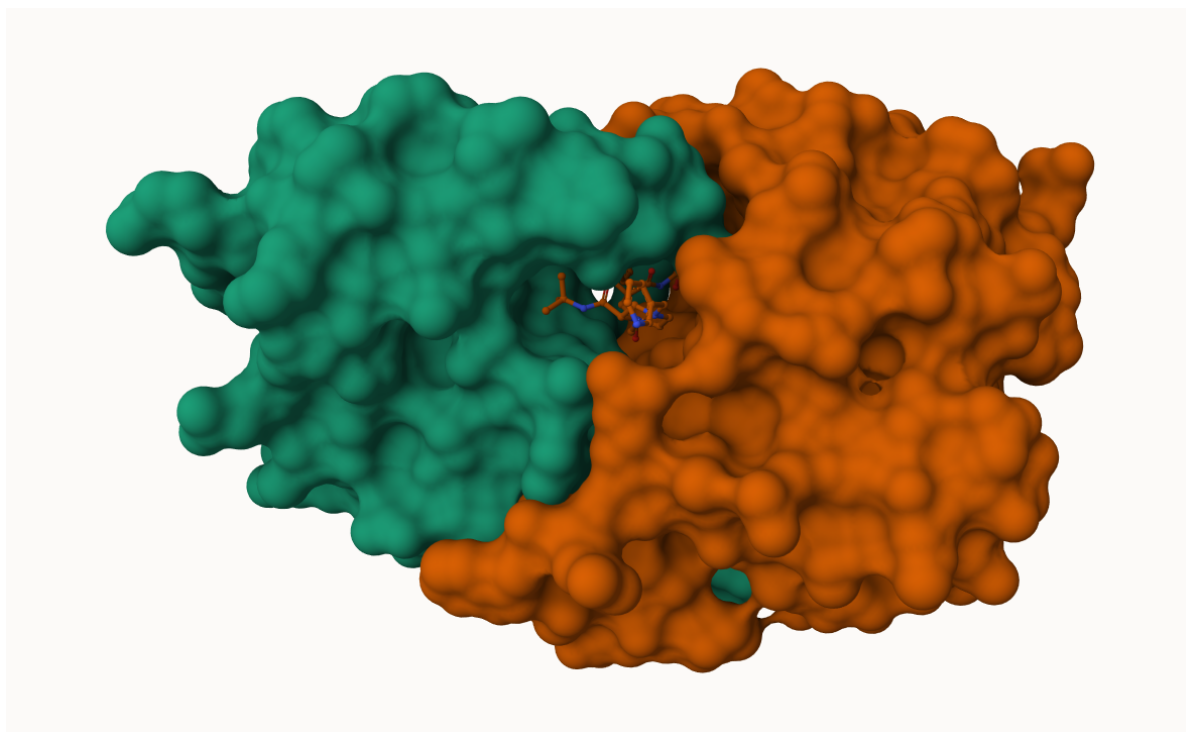


Figure 2: Surface display showing the binding cleft site for the inhibitor molecule



Figure 3: Spacefill display of catalytic ASP25 amino acids and key binding site water molecule

```
#install.packages("pak")

#pak::pkg_install(c("bioboot/bio3dview",
#                  "NGLVieweR",
#                  "bioc::msa"))
```

Getting started with the Bio3D package

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues in this PDB object.

Q8: Name one of the two non-protein residues?

One non-protein residue is HOH.

Q9: How many protein chains are in this structure?

There are 2 protein chains in this structure.

Bio3D is an R package from CRAN for structural bioinformatics.

```
library(bio3d)

pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87

```

6 ATOM      6      CG <NA>  PRO      A      1      <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>      N  <NA>
2  <NA>      C  <NA>
3  <NA>      C  <NA>
4  <NA>      O  <NA>
5  <NA>      C  <NA>
6  <NA>      C  <NA>

```

There are lots of functions that can work with these `pdb` objects:

```
head(pdbseq(pdb))
```

```

  1    2    3    4    5    6
"P" "Q" "I" "T" "L" "W"

```

```
library(bio3dview)
```

```
# view.pdb(pdb)
```

Let's try a custom view:

```
# view.pdb(pdb,colorScheme = "sse")
```

Q. Create a custom view of HIV-Pr highlighting the active site ASP (`resno=25`), the two chains (in your choice of colors) and the ligand all on a custom color background.

```

# active.site <- atom.select(pdb, resno=25)

# view.pdb(pdb,
#           cols = c("red", "blue"),
#           highlight = active.site,
#           highlight.style = "spacefill",
#           backgroundColor = "pink" )

```

Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given `pdb` object:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV  
TDELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

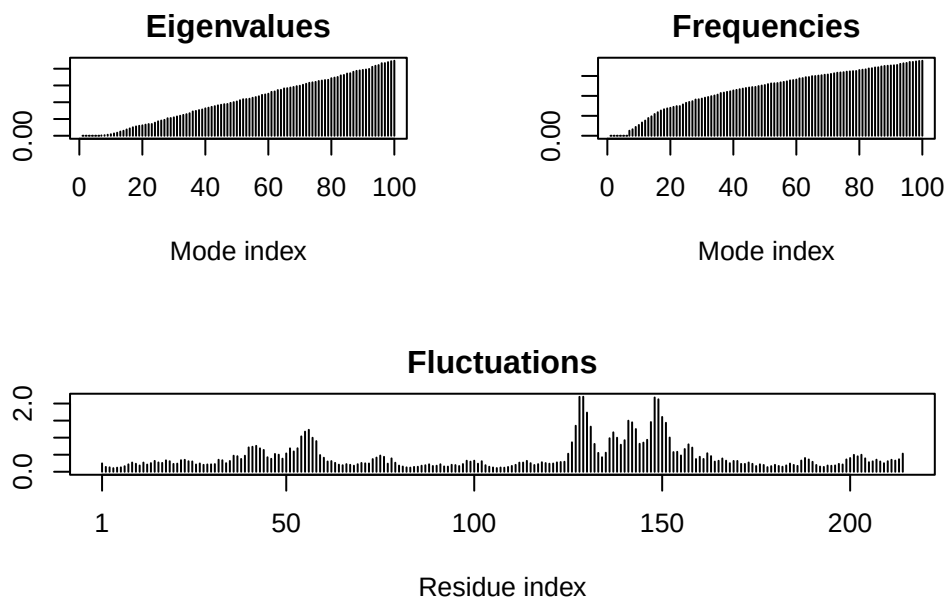
```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian... Done in 0.009 seconds.
```

```
Diagonalizing Hessian... Done in 0.169 seconds.
```

```
plot(m)
```



View the results

```
# view.nma(m)
```

Write out the results for viewing in Mol-star:

```
mktrj(m, file="nma.pdb")
```

Comparative analysis of the ADK family

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa

Q11. Which of the above packages is not found on BioConductor or CRAN?:

bio3dview

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

The sequence is 214 amino acids long.

Our first step is find a sequece for this family. WE will use the database ID “lake_A” here:

```
id <- "lake_A"
aa <- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMMLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

      121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTRKDDQEETVRKRLVEYHQMTAPLIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Search for related sequences in the database

```
blast <- blast.pdb(aa)
```

Searching ... please wait (updates every 5 seconds) RID = ZU49NU4M014

.....

Reporting 96 hits

```
head(blast$hit.tbl)
```

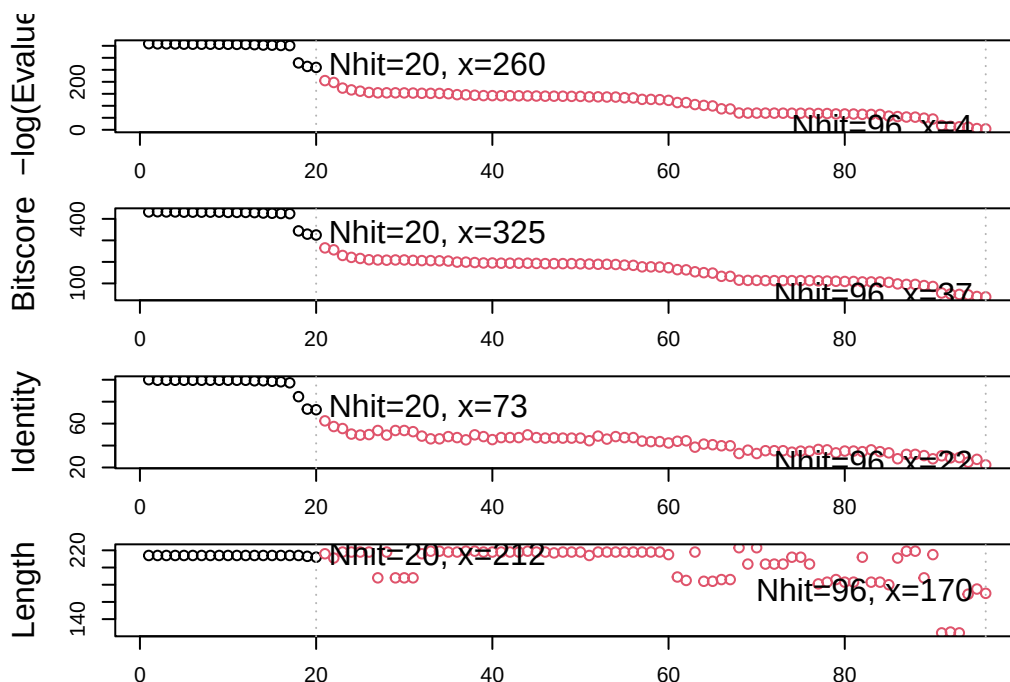
	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_7072423	1AKE_A	100.000	214	0	0	1
2	Query_7072423	8BQF_A	99.533	214	1	0	1
3	Query_7072423	4X8M_A	99.533	214	1	0	1
4	Query_7072423	6S36_A	99.533	214	1	0	1
5	Query_7072423	9R6U_A	99.533	214	1	0	1
6	Query_7072423	9R71_A	99.533	214	1	0	1

	q.end	s.start	s.end	evaluate	bitscore	positives	mlog.evaluate	pdb.id	acc
1	214	1	214	1.81e-156	432	100.00	358.6099	1AKE_A	1AKE_A
2	214	21	234	2.97e-156	433	100.00	358.1147	8BQF_A	8BQF_A
3	214	1	214	3.25e-156	432	100.00	358.0246	4X8M_A	4X8M_A
4	214	1	214	4.76e-156	432	100.00	357.6430	6S36_A	6S36_A
5	214	1	214	1.06e-155	431	99.53	356.8424	9R6U_A	9R6U_A
6	214	1	214	1.25e-155	431	99.53	356.6775	9R71_A	9R71_A

```
hits <- plot(blast)
```

* Possible cutoff values: 260 3
Yielding Nhits: 20 96

* Chosen cutoff value of: 260
Yielding Nhits: 20



```
hits$ pdb.id
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
[9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
files <- get.pdb(hits$ pdb.id, path="pdbc", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/6S36.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

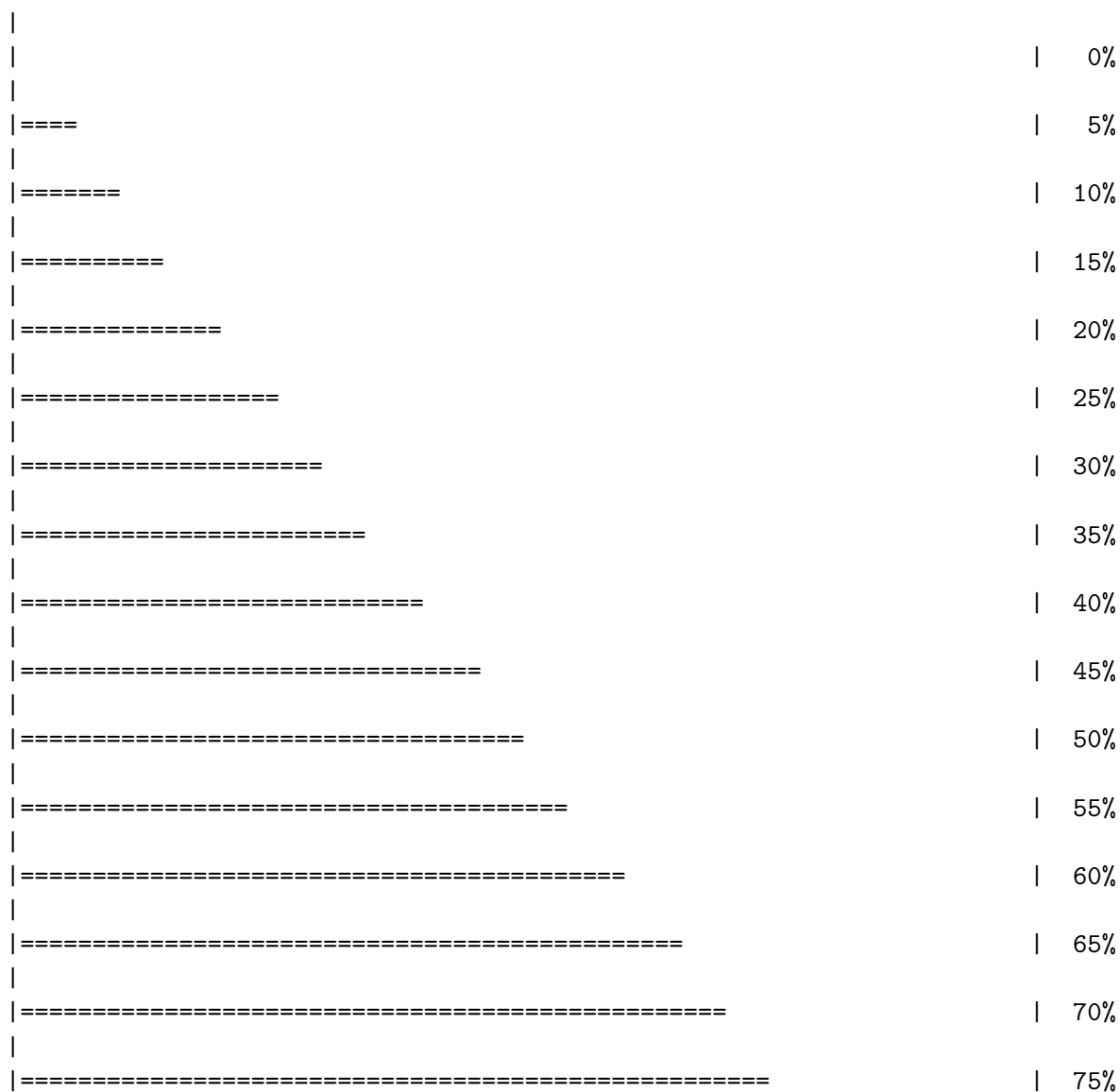
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download



```

|
|=====| 80%
|
|=====| 85%
|
|=====| 90%
|
|=====| 95%
|
|=====| 100%

```

Align and superpose all these ADK structures

```
pdbbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```

pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/8BQF_A.pdb
pdbbs/split_chain/4X8M_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/9R6U_A.pdb
pdbbs/split_chain/9R71_A.pdb
pdbbs/split_chain/8Q2B_A.pdb
pdbbs/split_chain/8RJ9_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/4X8H_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
pdbbs/split_chain/6HAM_A.pdb
pdbbs/split_chain/8PVW_A.pdb
pdbbs/split_chain/4K46_A.pdb
pdbbs/split_chain/4NP6_A.pdb

```

```

PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
.. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE

```

```

.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

pdb/seq: 1   name: pdbc/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbc/split_chain/8BQF_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbc/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbc/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbc/split_chain/9R6U_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbc/split_chain/9R71_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbc/split_chain/8Q2B_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbc/split_chain/8RJ9_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbc/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbc/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pdbc/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbc/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pdbc/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pdbc/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pdbc/split_chain/8PVW_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19  name: pdbc/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE

```

pdb/seq: 20 name: pdbs/split_chain/4NP6_A.pdb

pdbs

```

[Truncated_Name:1] 1AKE_A.pdb      1      .      .      .      40
[Truncated_Name:2] 8BQF_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:3] 4X8M_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:4] 6S36_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:5] 9R6U_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:6] 9R71_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:7] 8Q2B_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:8] 8RJ9_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:9] 6RZE_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:10] 4X8H_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:11] 3HPR_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:12] 1E4V_A.pdb      --MRIILLGAPVAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:13] 5EJE_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:14] 1E4Y_A.pdb      --MRIILLGALVAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:15] 3X2S_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:16] 6HAP_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:17] 6HAM_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:18] 8PVW_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:19] 4K46_A.pdb      --MRIILLGAPGAGKGTQAQFIMAKFGIPQISTGDMRLAA
[Truncated_Name:20] 4NP6_A.pdb      NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTGDMRLAA
                                *****  *****  *~*****
                                1      .      .      .      40

[Truncated_Name:1] 1AKE_A.pdb      41      .      .      .      80
[Truncated_Name:2] 8BQF_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:3] 4X8M_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQE---
[Truncated_Name:4] 6S36_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:5] 9R6U_A.pdb      VKSGSELGAQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:6] 9R71_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:7] 8Q2B_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:8] 8RJ9_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:9] 6RZE_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:10] 4X8H_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:11] 3HPR_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:12] 1E4V_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:13] 5EJE_A.pdb      VKSGSELGKQAKDIMDACKLVDELVIALVKERIAQEDCR
```

[Truncated_Name:14] 1E4Y_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR	
[Truncated_Name:15] 3X2S_A.pdb	VKSGSELGKQAKDIMDCGKLVDELVIALVKERIAQEDSR	
[Truncated_Name:16] 6HAP_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVRERICQEDSR	
[Truncated_Name:17] 6HAM_A.pdb	IKSGSELGKQAKDIMDAGKLVDEIIIALVKERICQEDSR	
[Truncated_Name:18] 8PVW_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR	
[Truncated_Name:19] 4K46_A.pdb	IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA	
[Truncated_Name:20] 4NP6_A.pdb	IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE	
	^* *^*** *** ^~* **~*~~~~~*~^*** *	
	41 . . .	80
	81 . . .	120
[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:2] 8BQF_A.pdb	-GFLLDGFPR TIPQADAMKEAGINVDYVIEFDVPDELIVD	
[Truncated_Name:3] 4X8M_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:4] 6S36_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:5] 9R6U_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:6] 9R71_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDALIVD	
[Truncated_Name:7] 8Q2B_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:8] 8RJ9_A.pdb	NGFLLAGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:9] 6RZE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:10] 4X8H_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:11] 3HPR_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:12] 1E4V_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:13] 5EJE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:14] 1E4Y_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:15] 3X2S_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:16] 6HAP_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:17] 6HAM_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:18] 8PVW_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:19] 4K46_A.pdb	KGFLLDGFPR TIPQADGLKEVGVVVDYVIEFDVADSVIVE	
[Truncated_Name:20] 4NP6_A.pdb	KGFLLDGFPR TIPQADGLKEMGINVDYVIEFDVADDVIVE	
	**** *****^~** *^ *****^***** * ^**^	
	81 . . .	120
	121 . . .	160
[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:2] 8BQF_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:3] 4X8M_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:4] 6S36_A.pdb	KIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:5] 9R6U_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:6] 9R71_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKA	
[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	

[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GETSGRV-----D
[Truncated_Name:19] 4K46_A.pdb	RMAGRRRAHLASGRTYHNVNPPKVEGKDDVTGEDLVIRED
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRRAHLPSGRTYHVNVNPPKVEGKDDVTGEDLVIRED
	^ * ***
	121 . . . 160
	161 . . . 200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQMTAALIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
	* * * * * ^ * * * * * * * * * *
	161 . . . 200
	201 . 216
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:2] 8BQF_A.pdb	TKPVAEVRADLEKIL-
[Truncated_Name:3] 4X8M_A.pdb	TKPVAEVRADLEKILG

```

[Truncated_Name:4] 6S36_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:5] 9R6U_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:6] 9R71_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:7] 8Q2B_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:8] 8RJ9_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:9] 6RZE_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:10] 4X8H_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:11] 3HPR_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:12] 1E4V_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:13] 5EJE_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:14] 1E4Y_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:15] 3X2S_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:16] 6HAP_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:17] 6HAM_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:18] 8PVW_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:19] 4K46_A.pdb     TKAVAEVSAELEKALA
[Truncated_Name:20] 4NP6_A.pdb     TKQVSEVSADIAKALA
                                ** * ** *~* *
                                201      .      216

```

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

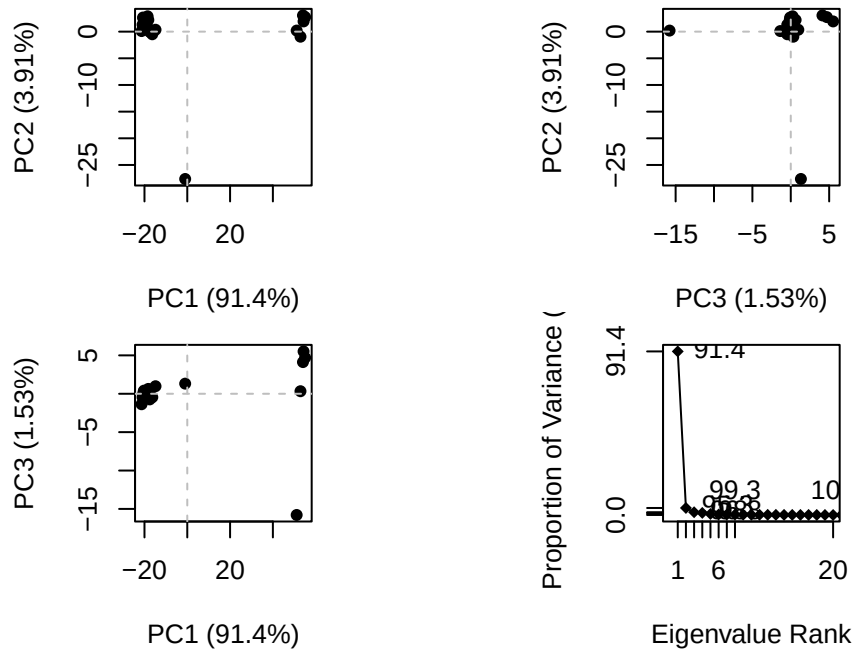
```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

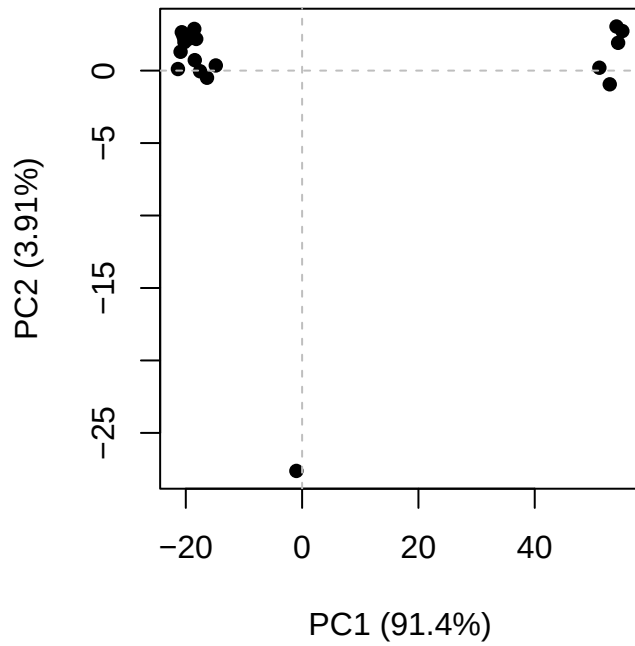
```
# view.pdbs(pdbs)
```

PCA of all this structural data (x,y,and z atom corrdinates):

```
pc <- pca(pdbs)
plot(pc)
```



```
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences:


```
# view.pca(pc)
```

```
mktrj(pc, file="pca.pdb")
```